

Quantum Field Theory and the Standard Model – Matthew D. Schwartz

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Contents

I	Field Theory	1
1	Historic Background	1
1.1	Blackbody Radiation	1
1.2	Second Quantization	2
1.2.1	Einstein Coefficients	2
1.2.2	Dirac	2

Part I

Field Theory

1 Historic Background

1.1 Blackbody Radiation

Blackbody: an object at some temperature (internal structure is irrelevant)

- A box with size L has EM waves s.t.

$$\lambda = L/n, \quad v = f\lambda$$
$$\omega_n = \frac{2\pi}{L} |\vec{n}|c$$

- **Classical equipartition theorem:** light is emitted equally in all modes. The intensity is proportional to the differential volume of phase space, thus by dimensional analysis,

$$I(\omega) \equiv \frac{1}{V} \frac{d}{d\omega} E(\omega) \longrightarrow I(\omega) = [c^{-3} \omega^2 k_B T]$$

We expect $I(\omega)$ to grow with angular frequency (ω), but this experimentally the graph is a Maxwell-Boltzmann distribution. Called the “ultraviolet catastrophe.”

Planck's solution: Energy in each EM mode is quantized.

$$E_n = \hbar\omega_n$$

- Interpretation: light is made up of massless particles (“photons”)
- If we calculate $I(\omega)$ using this, the catastrophe is averted.
- The derivation (see book) uses equilibrium stat mech – so light must “equilibrate,” i.e. particles must be created & destroyed. This is the domain of QFT.

$$I(\omega) = \frac{1}{V} \frac{dE(\omega)}{d\omega} = \frac{\hbar}{\pi^2} \frac{\omega^3}{e^{\hbar\omega\beta} - 1}$$

1.2 Second Quantization

1.2.1 Einstein Coefficients

Assume an atom with energy levels E_1, E_2 such that $E_2 - E_1 = \hbar\omega$. Further assume n_2 of the E_2 atoms and n_1 of the E_1 atoms.

A : Coefficient of spontaneous emission

B : Coefficient of stimulated emission

B' : Coefficient of absorption

A is the probability for a E_2 atom to become a E_1 atom by emitting a $\hbar\omega$ photon. B is the probability for an incoming $\hbar\omega$ photon to induce $E_2 \rightarrow E_1$. B' is the probability for an incoming photon to induce $E_1 \rightarrow E_2$.

We can estimate the differential dn_2 based on the above. Further, since particle number is conserved in this system,

$$\begin{aligned} dn_2 &\approx -[A + BI(\omega)] n_2 \\ dn_1 &= -dn_2 = -[A + BI(\omega)] n_2 + B'I(\omega)n_1 \end{aligned}$$

Using equilibrium stat mech, Einstein was able to predict the relation between A , B , and B' .

1.2.2 Dirac

Recall the quantum harmonic oscillator with creation/destruction operators $a^\dagger, a$.

$$\begin{aligned} [a, a^\dagger] &= 1 \\ \hat{N} &\equiv a^\dagger a, \quad \hat{N} |n\rangle = n |n\rangle \\ a^\dagger |n\rangle &= \sqrt{n+1} |n+1\rangle \\ a |n\rangle &= \sqrt{n} |n-1\rangle \end{aligned}$$

(Be able to derive above) Now we use Fermi's golden rule to calculate the transition rates.

$$\Gamma \sim |\mathcal{M}|^2 \delta(E_f - E_i), \quad \mathcal{M} = \langle f | H_{\text{int}} | i \rangle$$

H_{int} must have creation/destruction operators to create photons. It must also be Hermitian. Thus $H_{\text{int}} = H_I^\dagger a^\dagger + H_I a$. For the $2 \rightarrow 1$ transition, $|i\rangle = |\text{atom}_2\rangle \otimes |n_\omega\rangle$ and $|f\rangle = |\text{atom}_1\rangle \otimes |n_\omega + 1\rangle$ where n_ω (number of photons) has increased by one.

$$\begin{aligned}\mathcal{M}_{2 \rightarrow 1} &= \langle f | H_{\text{int}} | i \rangle \\ &= \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle \langle n_\omega + 1 | a^\dagger | n_\omega \rangle + \langle \text{atom}_1 | H_I | \text{atom}_2 \rangle \langle n_\omega + 1 | a | n_\omega \rangle \\ &= \mathcal{M}_0^\dagger \sqrt{n_\omega + 1}, \quad \mathcal{M}_0^\dagger \equiv \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle\end{aligned}$$

The $1 \rightarrow 2$ transition is very similar.

$$\mathcal{M}_{1 \rightarrow 2} = \mathcal{M}_0 \sqrt{n_\omega}, \quad \mathcal{M}_0 \equiv \langle \text{atom}_2 | H_I | \text{atom}_1 \rangle$$

Thus n_2 and n_1 change as

$$\begin{aligned}dn_2 &= -dn_1 = -|\mathcal{M}_{2 \rightarrow 1}|^2 n_2 + |\mathcal{M}_{1 \rightarrow 2}|^2 n_1 \\ &= -|\mathcal{M}_0|^2 (n_\omega + 1) n_2 + |\mathcal{M}_0|^2 (n_\omega) n_1\end{aligned}$$

Now we start the heavy calculations.

1. Energy is quantized by $\Delta = \hbar\omega = \hbar \frac{2\pi}{L} |\vec{n}|$ (with $c = 1$)

$$E(\omega) = \int^\omega (\hbar\omega) n_\omega d^3\vec{n} = (4\pi)\hbar L^3 \int_0^\omega \frac{d\omega}{(2\pi)^3} \omega^3 n_\omega$$

2. Multiply above by 2 for the 2 polarizations of light.

3. Intensity is

$$I(\omega) = \frac{1}{L^3} \frac{dE}{d\omega} = \frac{\hbar\omega^3}{\pi^2} n_\omega$$

So now we have

$$dn_2 = -dn_1 = -|\mathcal{M}_0|^2 \left[1 + \frac{\pi^2}{\hbar\omega^3} I(\omega) \right] + |\mathcal{M}_0|^2 \left[\frac{\pi^2}{\hbar\omega^3} I(\omega) \right] n_1$$

By comparison with the previous section, we read off the Einstein coefficients:

$$B' = B, \quad \frac{A}{B} = \frac{\hbar}{\pi^2} \omega^3$$